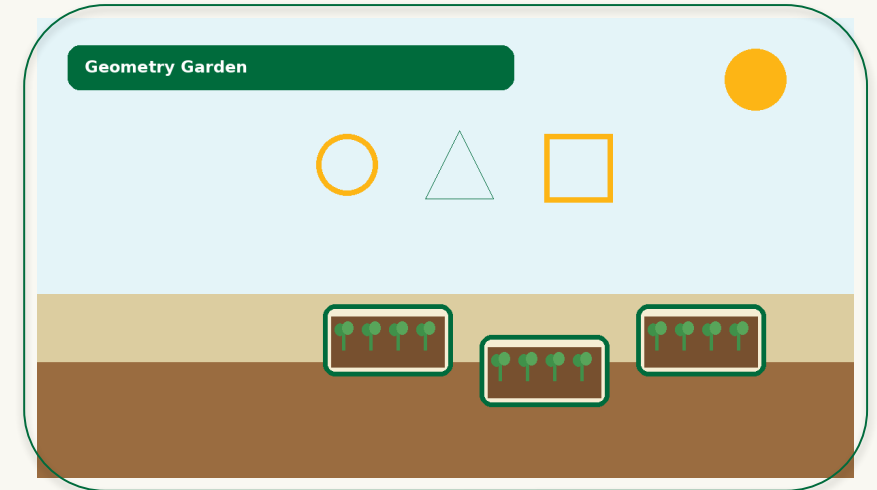


Geometry Garden

Mathematics in the Garden

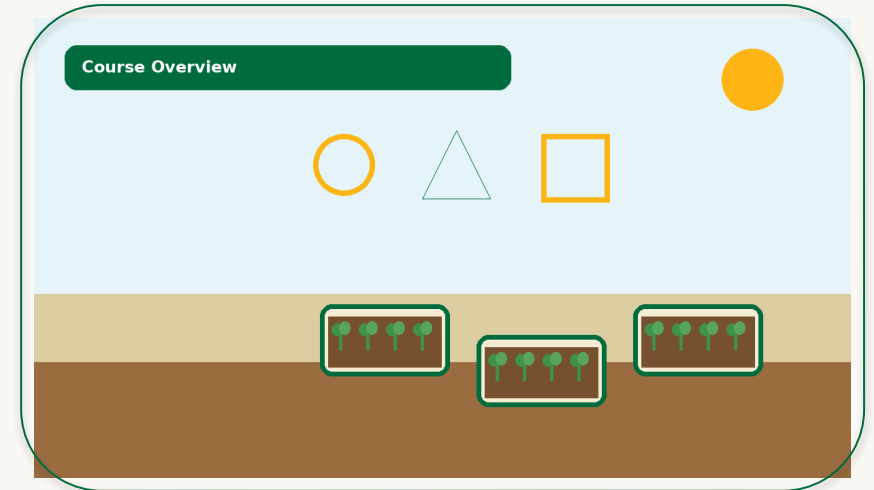
- LIFEWS AgriShine™ GrowMeal™ Primary 1-3 Teacher Training
- Audience: Primary 1-3 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised practical orientation
- Recommended duration: 3 hours online + 60-90 minutes garden practical



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden's educational and production purpose; teach the ten course topics accurately; connect practical gardening to Primary 1-3 curriculum; supervise age-appropriate garden work; maintain simple records; assess learning through practical and classroom evidence.
- Core materials: teacher guide, crop/word/number cards as relevant, child-safe hand tools, watering equipment, labels, clipboards, rulers/measuring tools where relevant, observation/record sheets, clean harvest containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe garden lesson, and submit a short lesson/assessment plan.



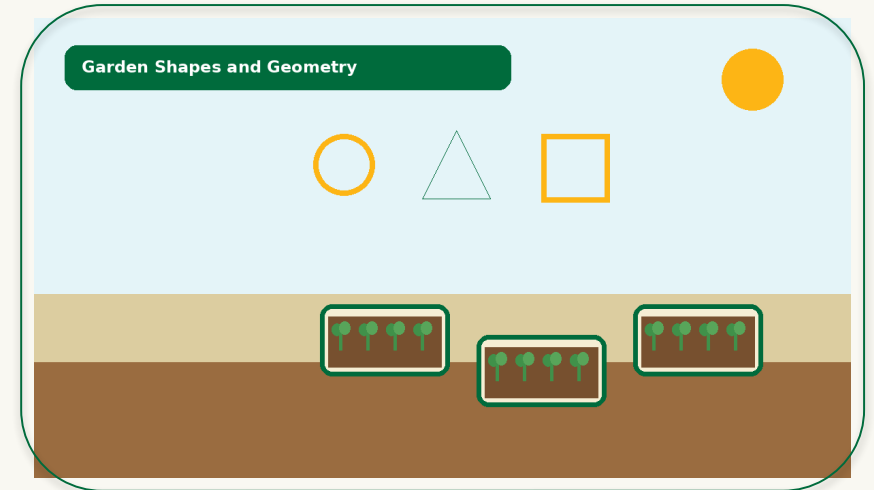
TEACHER PRACTICE

- Training sequence: EXPLAIN → DEMONSTRATE → PRACTISE → RECORD → REFLECT.

Garden Shapes and Geometry

Teacher knowledge and practical application

- Identify circles, triangles, squares, rectangles and other curriculum-appropriate shapes in beds and garden features.
- Distinguish the shape of a bed from the shape of objects inside it.
- Use large physical examples before drawings.
- Ask pupils to name and describe shapes using simple properties.
- Connect geometry language to the garden map.



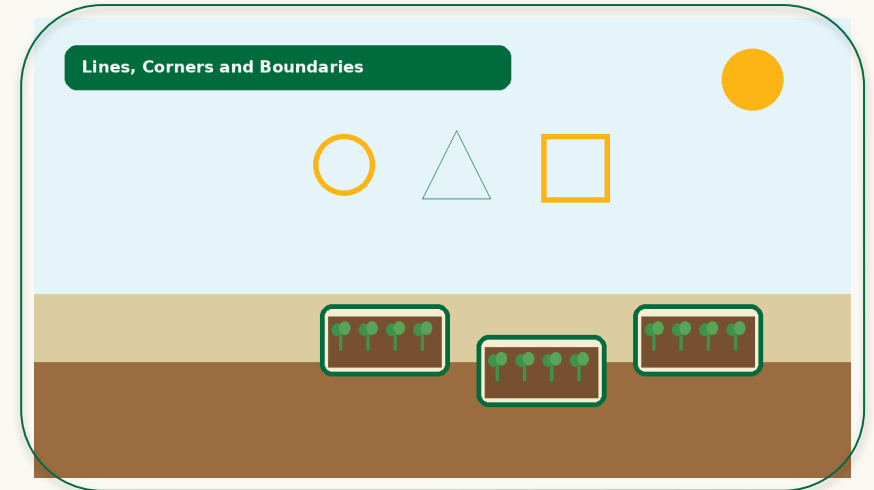
TEACHER PRACTICE

- Prepare a five-shape garden walk.

Lines, Corners and Boundaries

Teacher knowledge and practical application

- Identify straight and curved lines, sides, corners and boundaries.
- Trace bed edges visually or with safe walking routes.
- Use stakes/string only under teacher control and where safe.
- Compare shapes by number of sides and corners.
- Introduce boundary as the edge defining a growing area.



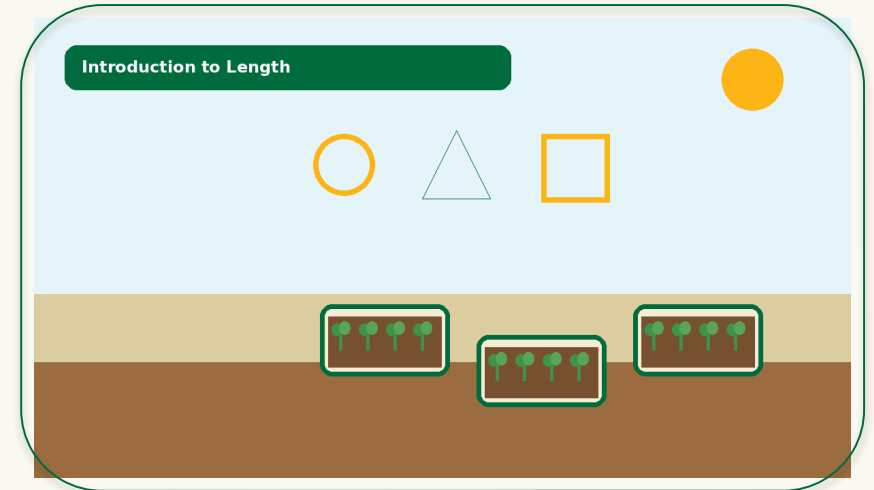
TEACHER PRACTICE

- Choose three beds for side-and-corner comparison.

Introduction to Length

Teacher knowledge and practical application

- Measure selected garden features with non-standard units and simple rulers/tapes.
- Teach where measurement starts and ends.
- Keep the measuring tool straight along the feature.
- Record number and unit together.
- Repeat a measurement to check consistency.



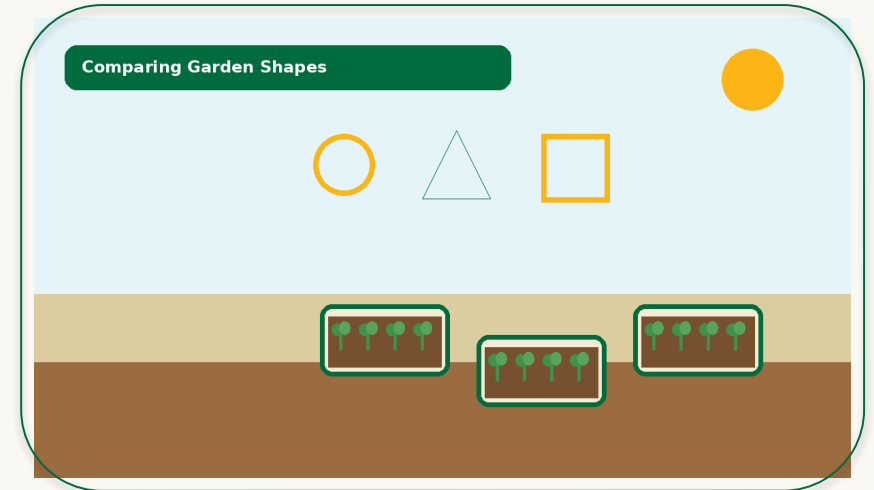
TEACHER PRACTICE

- Prepare one non-standard and one standard measurement activity.

Comparing Garden Shapes

Teacher knowledge and practical application

- Compare shapes by sides, corners, length or visual size without confusing different attributes.
- Ask pupils to state the feature they are comparing.
- Use matching and sorting before more formal descriptions.
- Discuss how two rectangles can look different yet remain rectangles.
- Use garden photographs or maps for follow-up work.



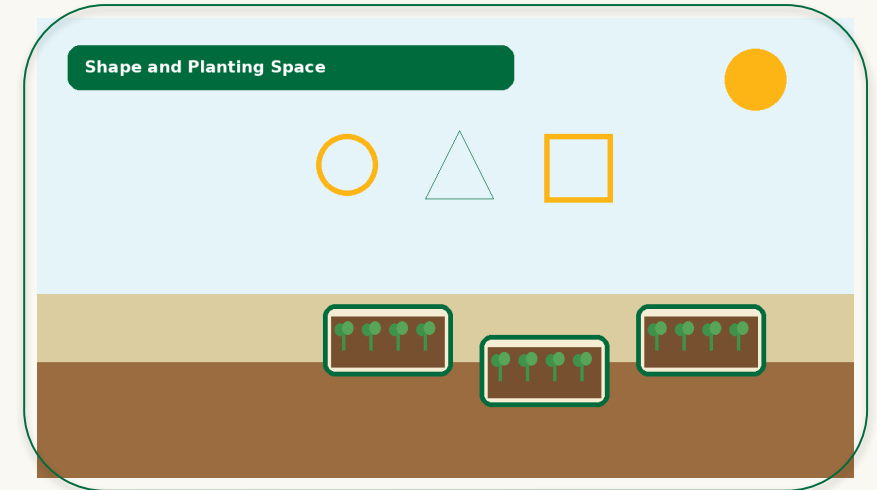
TEACHER PRACTICE

- Prepare a shape-sorting task based on garden features.

Shape and Planting Space

Teacher knowledge and practical application

- Show that bed shape influences how planting positions are arranged.
- Use dots or markers to plan planting positions inside a shape.
- Keep crop spacing agronomically appropriate.
- Compare how rows fit inside rectangular and circular spaces without claiming one is always better.
- Connect inside, outside, edge and center to planting layout.



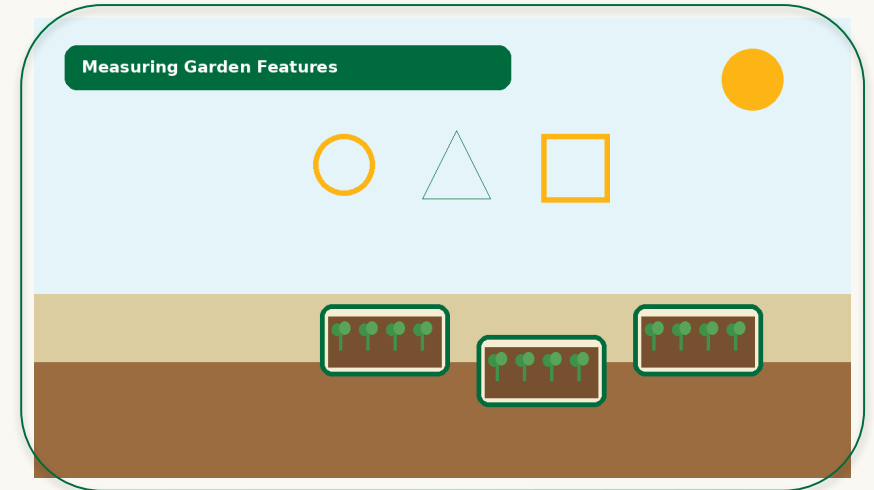
TEACHER PRACTICE

- Plan planting positions for two differently shaped beds.

Measuring Garden Features

Teacher knowledge and practical application

- Measure bed length, path width, container height or other safe features.
- Use pairs so one pupil positions the tool while another reads/records with supervision.
- Estimate before measuring where curriculum allows.
- Record units consistently.
- Avoid measuring near unsafe structures or active irrigation equipment.



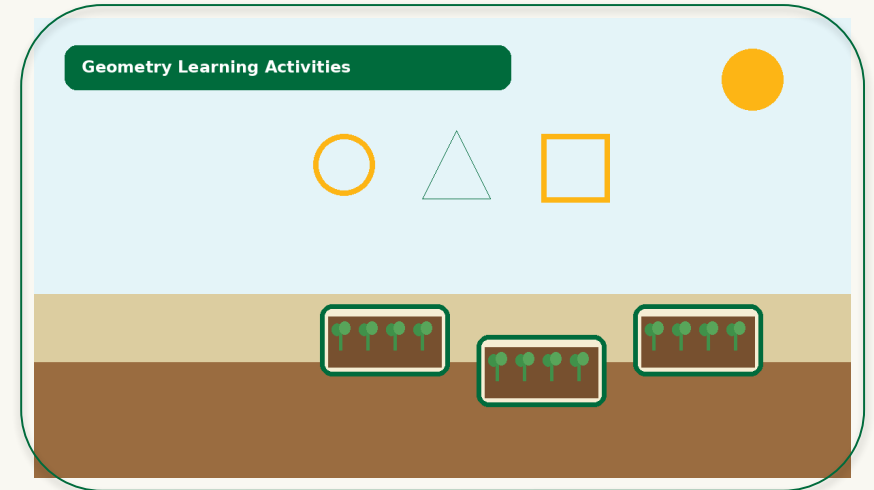
TEACHER PRACTICE

- Prepare a four-feature measurement sheet.

Geometry Learning Activities

Teacher knowledge and practical application

- Use shape hunts, bed-map matching, build-a-shape with cards, perimeter walks and design drawings.
- Keep physical movement organized along paths.
- Use manipulatives before abstract worksheets.
- Ask pupils to explain how they know a shape's name.
- Connect activities to actual garden maintenance or planting where possible.



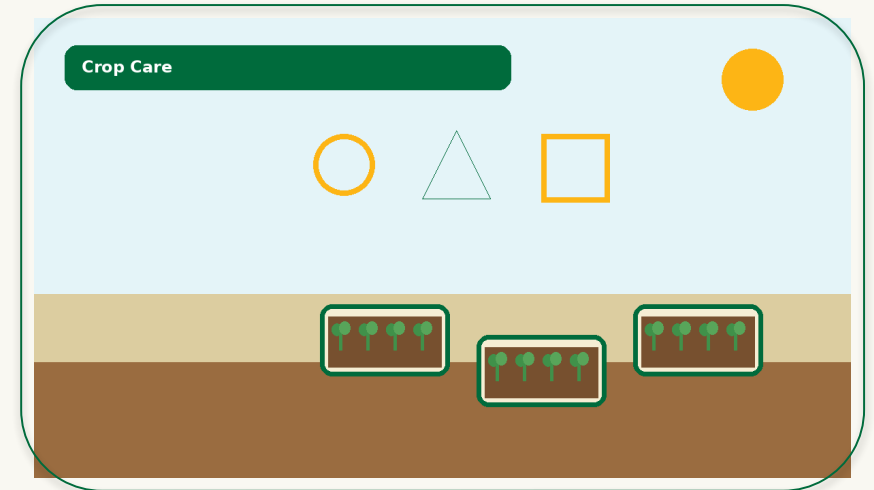
TEACHER PRACTICE

- Select two geometry stations for one lesson.

Crop Care

Teacher knowledge and practical application

- Integrate crop care without losing the mathematics objective.
- Count watering points, inspect bed boundaries and maintain clear paths.
- Avoid changing crop spacing merely to make a shape look perfect.
- Use care tasks to revisit position and measurement words.
- Record damaged edges or labels that affect teaching.



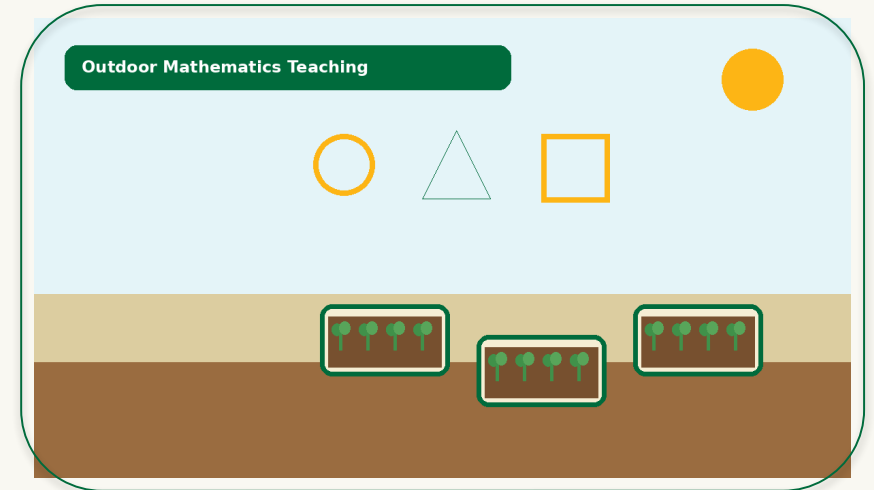
TEACHER PRACTICE

- Create a geometry-linked care routine.

Outdoor Mathematics Teaching

Teacher knowledge and practical application

- State the mathematics objective before entering the garden.
- Demonstrate the task at one station before group rotation.
- Use clipboards or durable recording sheets.
- Keep groups small and define where pupils may stand.
- Finish with a classroom-style summary using the measurements or shapes observed.



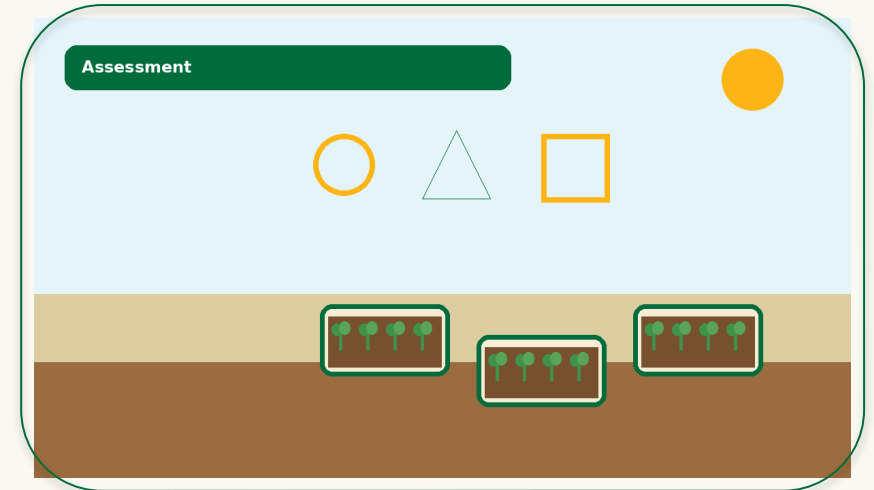
TEACHER PRACTICE

- Plan a 35-minute outdoor mathematics lesson.

Assessment

Teacher knowledge and practical application

- Assess shape recognition, properties, position language and basic measurement.
- Use practical identification and measuring tasks plus short recorded work.
- Check that pupils include units where required.
- Distinguish visual guessing from measured evidence.
- Use errors to decide which concept needs reteaching.



TEACHER PRACTICE

- Completion task: design and assess one Geometry Garden activity.